

TEST ITEM ANALYSIS REPORT

ACT 25MC1 - SCIENCE



Test Code: 25MC1

School: Sample High School

Grade: 11

Test Date: 10/31/2025

N* = 129

Number of Items = 34

Group: All Students Tested

ITEM	STRAND	BENCHMARK CODE		PERCENT CORRECT			A/F	B/G	C/H	D/J	E/K	BLANK	KEY	NGSS**
				0%	50%	100%								
IOD Interpretation of Data														
3	221	Identify basic features of a table, graph, or diagram in a simple presentation				5%	15%	9%	63%	8%	2%		D	
8	221	Identify basic features of a table, graph, or diagram in a simple presentation				8%	14%	9%	60%	9%	1%		J	
22	221	Identify basic features of a table, graph, or diagram in a simple presentation				6%	15%	7%	62%	9%	1%		J	
24	231	Find basic information in text that describes a simple data presentation				8%	14%	10%	60%	9%	0%		J	
37	231	Find basic information in text that describes a simple data presentation				6%	74%	15%	5%	0%	0%		B	
5	321	Understand basic scientific terminology				5%	15%	9%	63%	8%	2%		D	
12	341	Determine how the values of dependent variables change in a simple data presentation				6%	74%	14%	6%	0%	0%		G	
6	421	Translate information into a table, graph, or diagram				57%	22%	14%	7%	0%	0%		F	
39	421	Translate information into a table, graph, or diagram				10%	17%	61%	11%	0%	1%		C	
19	441	Perform a simple interpolation or simple extrapolation using data in a table or graph				5%	14%	9%	60%	9%	2%		D	
25	441	Perform a simple interpolation or simple extrapolation using data in a table or graph				5%	72%	15%	8%	0%	0%		B	
2	451	Compare or combine data from a simple data presentation (e.g., order or sum data from a table)				55%	22%	15%	8%	0%	0%		F	
1	541	Determine and/or use a simple mathematical relationship that exists between data				57%	22%	14%	7%	0%	0%		A	
4	541	Determine and/or use a simple mathematical relationship that exists between data				7%	17%	65%	10%	0%	1%		H	
15	541	Determine and/or use a simple mathematical relationship that exists between data				5%	72%	14%	9%	0%	0%		B	
16	541	Determine and/or use a simple mathematical relationship that exists between data				9%	18%	61%	12%	0%	1%		H	
27	541	Determine and/or use a simple mathematical relationship that exists between data				5%	15%	8%	61%	9%	2%		D	
28	541	Determine and/or use a simple mathematical relationship that exists between data				6%	70%	14%	10%	0%	0%		G	
38	541	Determine and/or use a simple mathematical relationship that exists between data				6%	71%	15%	8%	0%	0%		G	
40	541	Determine and/or use a simple mathematical relationship that exists between data				4%	15%	9%	62%	10%	1%		J	
26	551	Compare or combine data from two or more simple data presentations				5%	15%	8%	62%	9%	1%		J	
35	551	Compare or combine data from two or more simple data presentations				6%	72%	14%	8%	0%	0%		B	

Items 29, 30, 31, 32, 33 and 34 are field-tested and not included in the scoring.

Note: Percent may not equal 100% due to rounding error.

***Use the data with caution when N < 10.**

** Next Generation Science Standards (NGSS) are available at www.nextgenscience.org.

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CODE			0%	50%	100%								

Strand Percent Average Correct for 22 items = 64%

SIN Scientific Investigation

23	311	Understand the tools and functions of tools used in a complex experiment				24%	11%	63%	2%	0%	0%	C	
18	321	Understand the methods used in a simple experiment				57%	22%	13%	8%	0%	0%	F	
36	321	Understand the methods used in a simple experiment				58%	21%	14%	7%	0%	0%	F	
17	331	Find basic information in text that describes a complex experiment				6%	69%	16%	9%	0%	0%	B	
14	541	Determine the experimental conditions that would produce specified results				38%	19%	29%	12%	0%	2%	H	

Strand Percent Average Correct for 5 items = 55%

EMI Evaluation of Models, Inferences, and Experimental Results

7	311	Determine which models present certain basic information				26%	12%	58%	4%	0%	0%	C	
9	311	Determine which models present certain basic information				9%	19%	61%	11%	0%	1%	C	
10	311	Determine which models present certain basic information				57%	22%	13%	8%	0%	0%	F	
20	311	Determine which models present certain basic information				57%	22%	14%	7%	0%	0%	F	
21	311	Determine which models present certain basic information				23%	11%	63%	3%	0%	0%	C	
11	421	Identify similarities and differences between models				9%	18%	61%	11%	0%	1%	C	
13	431	Determine which simple hypothesis is, or is not, consistent with data or model				24%	11%	63%	2%	0%	0%	C	

Strand Percent Average Correct for 7 items = 60%

Overall Percent Average Correct for 34 items = 62%

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